Python 3

<table>
<tr><td>03</td><td>1</td><td><pre><code>Number = 0
while Number < 1 or Number > 10:
  Number = int(input("Enter a positive whole number: "))
  if Number > 10:
    print("Number too large")
  elif Number < 1:
    print("Not a positive number")
c = 1
for k in range(Number):
  print(c)
  c = (c * (Number - 1 - k)) // (k + 1)</code></pre></td><td>9</td></tr>
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<td>11</td>
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```
def SendMorseCode(MorseCode):
  PlainText = input("Enter your message (uppercase letters and
spaces only): ")
  PlainTextLength = len(PlainText)
  MorseCodeString = EMPTYSTRING
  for i in range(PlainTextLength):
    PlainTextLetter = PlainText[i]
    if PlainTextLetter == SPACE:
      Index = 0
    elif PlainTextLetter >= 'A' and PlainTextLetter <= 'Z':
      Index = ord(PlainTextLetter) - ord('A') + 1
    else:
      ReportError("Invalid character entered")
      Index = 0
      MorseCodeString = EMPTYSTRING
      break
    CodedLetter = MorseCode[Index]
    MorseCodeString = MorseCodeString + CodedLetter + SPACE
  print(MorseCodeString)
```

Alternative answer:

```
def SendMorseCode(MorseCode):
  PlainText = input("Enter your message (uppercase letters and
spaces only): ")
  Valid = True
  for Character in PlainText:
    if Character != SPACE:
      if Character < "A" or Character > "Z":
        Valid = False
        MorseCodeString = EMPTYSTRING
        ReportError("Invalid character entered")
        break
  if Valid:
    PlainTextLength = len(PlainText)
    MorseCodeString = EMPTYSTRING
    for i in range(PlainTextLength):
      PlainTextLetter = PlainText[i]
      if PlainTextLetter == SPACE:
        Index = 0
      else:
        Index = ord(PlainTextLetter) - ord('A') + 1
      CodedLetter = MorseCode[Index]
      MorseCodeString = MorseCodeString + CodedLetter + SPACE
  print(MorseCodeString)
```

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12	**1**	<pre>def SendSignals(MorseCodeString):
		  Transmission = EMPTYSTRING
		  CodeStringLength = len(MorseCodeString)
		  for i in range(CodeStringLength):
		    Symbol = MorseCodeString[i]
		    if Symbol == '.':
		      SymbolString = "= "
		    elif Symbol == '-':
		      SymbolString = "=== "
		    elif Symbol == SPACE:
		      SymbolString = SPACE + SPACE
		    Transmission = Transmission + SymbolString
		  print(Transmission)</pre>
			7

13 **1**

```
def OutputAlphabetWithCode(Letter, MorseCode):
  for Ptr in range(1, 27):
    print(Letter[Ptr], end=SPACE)
    print('{0:<5}'.format(MorseCode[Ptr]), end=SPACE)
    if Ptr % 4 == 0:
      print()
  print()

def DisplayMenu():
  print()
  print("Main Menu")
  print("=========")
  print("R - Receive Morse code")
  print("S - Send Morse code")
  print("A - Output alphabet with Morse code")
  print("X - Exit program")
  print()

def SendReceiveMessages():
Dash = [20,23,0,0,24,1,0,17,0,21,0,25,0,15,11,0,0,0,0,22,13,0,0,10,0,0,0]
  Dot = [5,18,0,0,2,9,0,26,0,19,0,3,0,7,4,0,0,0,12,8,14,6,0,16,0,0,0]
  Letter =
["SPACE",'A','B','C','D','E','F','G','H','I','J','K','L','M','N','O','P','Q

  MorseCode = ["SPACE",'.-','-...','-.-.','-..','.','..-.','--.','....','..
','.--.','--.-','.-.','...','-','..-','...-','.--','-..-','-.--','--..']
  ProgramEnd = False
  while not ProgramEnd:
    DisplayMenu()
    MenuOption = GetMenuOption()
    if MenuOption == 'R':
      ReceiveMorseCode(Dash, Letter, Dot)
    elif MenuOption == 'S':
      SendMorseCode(MorseCode)
    elif MenuOption == 'A':
      OutputAlphabetWithCode(Letter, MorseCode)
    elif MenuOption == 'X':
      ProgramEnd = True
```

14 **1**

```
def SendMorseCode(MorseCode, Keys):
  PlainText = input("Enter your message (uppercase letters and spaces only)
  PlainTextLength = len(PlainText)
  MorseCodeString = EMPTYSTRING
  for i in range(PlainTextLength):
    PlainTextLetter = PlainText[i]
    if PlainTextLetter == SPACE:
      Index = 0
    else:
      Index = ord(PlainTextLetter) - ord('A') + 1
    Index = Index + Keys[i % 3]
    while Index < 0:
      Index = Index + 27
    while Index >= 27:
      Index = Index - 27
    CodedLetter = MorseCode[Index]
    MorseCodeString = MorseCodeString + CodedLetter + SPACE
  print(MorseCodeString)

def SendReceiveMessages():
Dash = [20,23,0,0,24,1,0,17,0,21,0,25,0,15,11,0,0,0,0,22,13,0,0,10,0,0,0]
  Dot = [5,18,0,0,2,9,0,26,0,19,0,3,0,7,4,0,0,0,12,8,14,6,0,16,0,0,0]
  Letter = 
["SPACE",'A','B','C','D','E','F','G','H','I','J','K','L','M','N','O','P','Q

  MorseCode = ["SPACE",'.-','-...','-.-.','-..','.','..-.','--.','....','..
','.--.','--.-','.-.','...','-','..-','...-','.--','-..-','-.--','--..']
  ProgramEnd = False

  Keys = [0,0,0]
  for i in range(3):
    ValidDisplacement = False
    while not ValidDisplacement:
      try:
        Displacement = int(input("Enter encryption key (integer): "))
        ValidDisplacement = True
      except:
        pass
    Keys[i] = Displacement

  while not ProgramEnd:
    DisplayMenu()
    MenuOption = GetMenuOption()
    if MenuOption == 'R':
      ReceiveMorseCode(Dash, Letter, Dot)
    elif MenuOption == 'S':
      SendMorseCode(MorseCode, Keys)
    elif MenuOption == 'X':
      ProgramEnd = True
```

Alternative answer:

```
def SendMorseCode(MorseCode, Key1, Key2, Key3):
  PlainText = input("Enter your message (uppercase letters and spaces only)
  PlainTextLength = len(PlainText)
  MorseCodeString = EMPTYSTRING
  for i in range(PlainTextLength):
    PlainTextLetter = PlainText[i]
    if PlainTextLetter == SPACE:
      Index = 0
    else:
      Index = ord(PlainTextLetter) - ord('A') + 1
    if i % 3 == 0:
      Displacement = Key1
    elif i % 3 == 1:
      Displacement = Key2
    else:
      Displacement = Key3
    Index = Index + Displacement
    while Index < 0:
      Index = Index + 27
    while Index >= 27:
      Index = Index - 27
    CodedLetter = MorseCode[Index]
    MorseCodeString = MorseCodeString + CodedLetter + SPACE
  print(MorseCodeString)

def SendReceiveMessages():
Dash = [20,23,0,0,24,1,0,17,0,21,0,25,0,15,11,0,0,0,0,22,13,0,0,10,0,0,0]
  Dot = [5,18,0,0,2,9,0,26,0,19,0,3,0,7,4,0,0,0,12,8,14,6,0,16,0,0,0]
  Letter =
["SPACE",'A','B','C','D','E','F','G','H','I','J','K','L','M','N','O','P','Q

  MorseCode = ["SPACE",'.-','-...','-.-.','-..','.','..-.','--.','....','..
','.--.','--.-','.-.','...','-','..-','...-','.--','-..-','-.--','--..']
  ProgramEnd = False

  ValidDisplacement = False
  while not ValidDisplacement:
    try:
      Key1 = int(input("Enter encryption key (integer): "))
      ValidDisplacement = True
    except:
      pass
  ValidDisplacement = False
  while not ValidDisplacement:
    try:
      Key2 = int(input("Enter encryption key (integer): "))
      ValidDisplacement = True
    except:
      pass
  ValidDisplacement = False
```

```
while not ValidDisplacement:
  try:
    Key3 = int(input("Enter encryption key (integer): "))
    ValidDisplacement = True
  except:
    pass

while not ProgramEnd:
  DisplayMenu()
  MenuOption = GetMenuOption()
  if MenuOption == 'R':
    ReceiveMorseCode(Dash, Letter, Dot)
  elif MenuOption == 'S':
    SendMorseCode(MorseCode, Key1, Key2, Key3)
  elif MenuOption == 'X':
    ProgramEnd = True
```